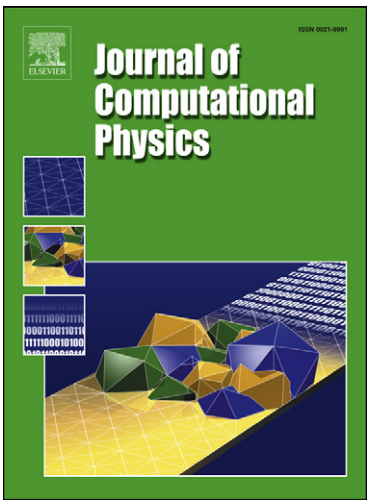
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A family of high-order targeted ENO scheme for compressible-fluid simulations

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Abstract

Although classical WENO schemes have achieved great success and are widely accepted, they exhibit several shortcomings. They are too dissipative for direct simulations of turbulence and lack robustness when very-high-order versions are applied to complex flows. In this paper, we propose a family of high-order targeted ENO schemes which are applicable for compressible-fluid simulations involving a wide range of flow scales. In order to increase the numerical robustness as compared to very-high-order classical WENO schemes, the reconstruction dynamically assembles a set of low-order candidate stencils with incrementally increasing width. While discontinuities and small-scale fluctuations are efficiently separated, the numerical dissipation is significantly diminished by an ENO-like stencil selection, which either applies a candidate stencil with its original linear weight, or removes its contribution when it is crossed by a discontinuity. The background linear scheme is optimized under the constraint of preserving an approximate dispersion-dissipation relation. By means of quasi-linear analyses and practical numerical experiments, a set

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of case-independent parameters is determined. The general formulation of arbitrarily high-order schemes is presented in a straightforward way. A variety of benchmark-test problems, including broadband waves, strong shock and contact discontinuities are studied. Compared to well-established classical WENO schemes, the present schemes exhibit significantly improved robustness, low numerical dissipation and sharp discontinuity capturing. They are particularly suitable for DNS and LES of shock-turbulence interactions.

Keywords: WENO-JS, WENO-Z, WENO-CU6, optimal scheme, high-order scheme, shock-turbulence interaction

1. Introduction

The development of advanced numerical schemes, that are simultaneously capable of resolving small-scale flow structures and of capturing discontinuities, is a long-standing subject of computational fluid dynamics research. The apparent contradicting requirements of low-dissipation for resolving small-scale flow features and sufficient numerical dissipation for stable capturing of discontinuities pose the key problem [1]. Several approaches, such as total variation diminishing (TVD) schemes [2], essentially non-oscillatory (ENO) [3] and weighted essentially non-oscillatory (WENO) schemes [4], have been presented in the paper to overcome this problem. While TVD schemes are able to provide sufficient numerical dissipation to capture discontinuities, and to restore formal high-order accuracy in smooth regions without critical points [5][6][7], they degenerate to first-order and produce excessive numerical dissipation near critical points. ENO schemes choose the smoothest stencil to capture discontinuities from a set of candidate approximation stencils

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and reduce numerical dissipation compared to TVD schemes [3][8]. WENO schemes exploit a weighted average of approximations from all candidate stencils. Based on the smoothness indicators, the weights are designed to recover the ENO property for capturing discontinuities and to restore the background linear schemes in smooth regions of the solution [4][9]. WENO schemes are widely used in gas dynamics research [10][11][12][13]. However, for turbulent flow simulations classical WENO schemes exhibit excessive dissipation, which damps the small-scale flow structures significantly [14][15]. As shown by the approximated dispersion relation (ADR) analysis [16], the spectral property of WENO schemes is inferior to that of low-dissipation linear schemes. Furthermore, although the classical fifth-order WENO scheme is sufficiently robust for many applications, even higher-order schemes may fail when there exist multiple discontinuities close to each other [17][18].

Noticing that classical WENO schemes fail to recover the formal fifth-order accuracy near critical points, where low-order derivatives vanish, Henrick et al. [19] propose WENO-M to deal with this problem by remapping the WENO-JS weights [9] so that the sufficient criteria for fifth-order are satisfied. Borges et al. [20] find that the improved results of WENO-M for shock capturing problems are due to the non-linear adaptation, i.e. larger weights assigned to the non-smooth stencils, rather than due to the high-order accuracy in smooth regions. In order to decrease the dissipation induced by the nonlinear adaption, Hill and Pullin [21] propose a switch function to freeze the adaption when the ratio between the largest and the smallest calculated smoothness indicator does not exceed a certain threshold. However, the threshold value for this shifting is likely to be problem dependent,

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i.e. large thresholds may cause numerical problems whereas small values degenerate the effectiveness [22]. In order to achieve both high order and less nonlinear adaption, Hu et al. [23] propose an adaptive central-upwind 6th-order WENO-CU6 scheme by introducing the contribution from a downwind stencil. WENO-CU6 has been further developed into an Implicit Large Eddy Simulation (ILES) approach by incorporating a physically-motivated scale-separation formulation [24]. The drawback of WENO-CU6 schemes is that spurious waves may appear at marginally-resolved wave numbers as the background central scheme cannot suppress disturbances from accumulated dispersion errors [25].

The built-in dissipation from the background linear scheme is another main dissipation source [22][26][27]. One choice is to optimize directly the spectral property of the background linear scheme. Weirs and Candler [28] construct a six-point scheme with low dissipation and third-order accuracy by introducing a downwind stencil. Another approach [22] is to optimize stencil weights by minimizing the dispersion and dissipation errors simultaneously and consequentially delaying the dissipative bandwidth to a certain wave number. However, the resulting optimal background schemes typically exhibit some anti-dissipation for a certain wave-number range [22]. Hu et al. [25] propose a general dispersion-dissipation relation, which can serve as a guideline for optimizing the dispersion and dissipation of both linear and nonlinear schemes. The minimum dissipation is determined such that it is sufficient to damp spurious high wave-number oscillations. **Numerical experiments** show that the optimized schemes perform well for turbulence simulations [29][25].

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Several methods have been developed to improve the numerical robustness of very high-order WENO schemes. Suresh and Huynh [30] propose to bound the solution near discontinuities by a monotonicity-preserving method. The problem of this technique is that it fails to maintain the formal order of the original WENO scheme [17]. Another approach is the order-reduction method [17][18], which recursively resorts to lower-order stencils. However, feasibility of the high-order approximation is checked by an empirical reconstruction-failure detector based on density and pressure, which makes it problematic for complex applications. Recently, an *a posteriori* approach, which first detects the critical numerical fluxes that may lead to negative pressure or density and then imposes a positivity-preserving flux limiter to correct the fluxes, has been proposed by Hu et al. [31]. This approach is simple and effective for general high-order conservative schemes.

In this paper, we propose a family of high-order targeted ENO (TENO) schemes preserving low dissipation for the low to intermediate wave-number range, sharp shock-capturing capacity and good robustness. Several new ideas are introduced to resolve the aforementioned issues:

1. Different from the classical WENO scheme assembling candidate stencils of same width, we create a high-order reconstruction by assembling a set of low-order (higher than third) upwind-biased candidate stencils with incremental width. The new scheme can gradually degenerate to third order, according to local flow features, thus avoiding the problem of multiple discontinuities and recovering the robustness of the classical fifth-order WENO-JS scheme.
2. Inspired by the work of Hu and Adams [24], an even stronger scale-

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separation formulation is designed to isolate discontinuities from high wave-number physical fluctuations.

3. Unlike WENO which assigns smooth weights to candidate stencils, a candidate stencil either is applied with optimal weight, or eliminated entirely when it contains a genuine discontinuity with certain strength, following on ENO-like stencil selection. This approach suppresses numerical dissipation to the limit of the background linear scheme for the wave-number region of interest while it preserves robust shock-capturing capability.
4. Since the background linear scheme is one main source of numerical dissipation, the optimization algorithm proposed by Hu and Adams [25] is adopted. Such optimization results in an order degeneration of one, but provides favourable spectral properties satisfying the dispersion-dissipation relation.

It is straightforward and fairly simple to construct arbitrarily high-order TENO schemes. As canonical example, we focus on fifth-order and sixth-order TENO schemes and discuss their performances for a wide range of test problems. The remainder of this paper is organized as follows. In section 2, a brief summary of WENO schemes is given as general introduction. In section 3, generalized formulations of the new TENO schemes are proposed, and the construction of fifth-order and sixth-order TENO schemes is discussed in detail. In section 4, an optimization algorithm is introduced for fifth-order and sixth-order TENO schemes in combination with the ADR analysis. In section 5, a set of benchmark cases involving strong discontinuities and broadband fluctuations is simulated. The performance of the proposed schemes for com-

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pressible flow simulations is discussed and compared with that of established WENO schemes. In the last section, the overall properties of the proposed schemes are discussed.

2. Brief review of WENO schemes

In this section, we first review basic concepts of classical WENO schemes. Then the construction of typical five-point and six-point stencil schemes is given.

2.1. Fundamental concepts of WENO schemes

For simplicity, we consider the one-dimensional hyperbolic conservation law

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} f(u) = 0, \tag{1}$$

where the characteristic velocity is further assumed to be positive $\frac{\partial f(u)}{\partial u} > 0$ without loss of generality. After the discretization of Eq. (1) on a uniform mesh, e.g. x_i, a system of ordinary differential equations

$$\frac{du_i}{dt} = -\frac{\partial f}{\partial x}\,|_{x=x_i}\ ,\ i = 0, \cdots, n, \tag{2}$$

is formed and marched in time by an explicit strongly-stable Runge-Kutta method. Eq. (1) can be approximated by a conservative finite difference scheme in semi-discrete form as

$$\frac{du_i}{dt} = -\frac{1}{\Delta x}(h_{i+1/2} - h_{i-1/2}), \tag{3}$$

where the primitive function $h(x)$ is implicitly defined by

$$f(x) = \frac{1}{\Delta x}\int_{x-\Delta x/2}^{x+\Delta x/2} h(\xi)d\xi. \tag{4}$$

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Eq. (3) is further approximated as

$$\frac{du_i}{dt} \approx -\frac{1}{\Delta x}(\widehat{f}_{i+1/2} - \widehat{f}_{i-1/2}), \tag{5}$$

where the numerical fluxes $\hat{f}_{i\pm1/2}$ is computed from a convex combination of r candidate-stencil fluxes

$$\widehat{f}_{i+1/2} = \sum_{k=0}^{r-1} w_k \widehat{f}_{k,i+1/2}. \tag{6}$$

To obtain a $(2r-1)$-order approximation for $\widehat{f}_{i+1/2}$, a $(r-1)$-degree interpolation on each candidate stencil leads to

$$h(x) \approx \hat{f}_k(x) = \sum_{l=0}^{r-1} a_{l,k} x^l. \tag{7}$$

After substituting Eq. (7) into Eq. (4) and evaluating the integral functions at the stencil nodes, the coefficients a_l are determined by solving the resulting system of linear algebraic equations. The negative split flux can be obtained by symmetry at the cell interface $x_{i+1/2}$.

2.2. Fifth-order WENO-JS and WENO-Z schemes

For the classical fifth-order WENO scheme with $r = 3$, a 2-degree polynomial approximation results in

$$h(x) \approx \widehat{f}_k(x) = a_{0,k} + a_{1,k}x + a_{2,k}x^2. \tag{8}$$

For each stencil, a 3rd-order approximation of the numerical flux evaluated at the cell face $i + \frac{1}{2}$ gives

$$\begin{aligned}
\hat{f}_{0,i+1/2} &= \tfrac{1}{6}(2f_{i-2} - 7f_{i-1} + 11f_i), \\
\hat{f}_{1,i+1/2} &= \tfrac{1}{6}(-f_{i-1} + 5f_i + 2f_{i+1}), \\
\hat{f}_{2,i+1/2} &= \tfrac{1}{6}(2f_i + 5f_{i+1} - f_{i+2}).
\end{aligned} \tag{9}$$

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With Taylor expansion analysis, we obtain

$$\hat{f}_{k,i+1/2} = h_{k,i+1/2} + A_k \Delta x^3 + \mathcal{O}(\Delta x^4),\ k = 0, 1, 2, \tag{10}$$

where A_k is a constant independent of Δx and may be different for different candidate stencils. For suppressing Gibbs oscillations [32] in the presence of discontinuities, the weights in Eq. (6) are defined as

$$w_k = \frac{\alpha_k}{\sum_{k=0}^{2} \alpha_k},\ \alpha_k = \frac{d_k}{(\beta_k + \varepsilon)^q}, \tag{11}$$

where d_k are optimal weights that generate a $(2r-1)$-th order scheme on a $(2r-1)$-point full stencil. Here, $d_0 = 0.1, d_1 = 0.6, d_2 = 0.3$. ε is a small value to prevent division by zero, whereas further research has shown that it indeed acts as a cutoff of the smoothness measure [33]. $q \geq 1$, generally defined as $q = 2$, is devised to accelerate the nonlinear adaptation towards the essentially non-oscillatory property. The local smoothness indicator β_k, determining to what extent certain candidate stencil should contribute to the final reconstruction, is defined as

$$\beta_k = \sum_{j=1}^{2} \Delta x^{2j-1} \int_{x_{i-1/2}}^{x_{i+1/2}} \left(\frac{d^j}{dx^j} \widehat{f}_k(x) \right)^2 dx \tag{12}$$

according to Jiang and Shu [9]. Explicit expression in terms of cell-averaged quantities f_i gives

$$\begin{aligned}
\beta_0 &= \tfrac{1}{4}(f_{i-2} - 4f_{i-1} + 3f_i)^2 + \tfrac{13}{12}(f_{i-2} - 2f_{i-1} + f_i)^2, \\
\beta_1 &= \tfrac{1}{4}(f_{i-1} - f_{i+1})^2 + \tfrac{13}{12}(f_{i-1} - 2f_i + f_{i+1})^2, \\
\beta_2 &= \tfrac{1}{4}(3f_i - 4f_{i+1} + f_{i+2})^2 + \tfrac{13}{12}(f_i - 2f_{i+1} + f_{i+2})^2.
\end{aligned} \tag{13}$$

In smooth regions, the local variation of the smoothness indicators is $\mathcal{O}(\Delta x^4)$, leading to w_k with approximately optimal weight. The convex combination of

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r stencils produces a $(2r-1)$-th order numerical flux approximation at the cell face. If a certain stencil k contains a discontinuity, β_k is $\mathcal{O}(1)$ while the others are $\mathcal{O}(\Delta x^2)$. The corresponding weight w_k becomes very small, thus the discontinuity is avoided, but the scheme degenerates to r-th order. Henrick et al. [19] investigate the effective order of classical WENO schemes and conclude that an overall $(2r-1)$-th order accuracy requires the condition $w_k - d_k = \mathcal{O}(\Delta x^3)$ to be satisfied in smooth regions. WENO-JS [9] degenerates to third-order near critical points, where the first-order derivative vanishes. Only second-order overall accuracy can be achieved when the second-order derivative vanishes as well. Motivated by the work of Henrick et al. [19], Borges et al. [20] propose a new weighting strategy as

$$\alpha_k = d_k \left(1 + \left(\frac{\tau_5}{\beta_k + \varepsilon}\right)^q\right),\ \tau_5 = |\beta_0 - \beta_2|\,, \tag{14}$$

by introducing a high-order reference value calculated from the full $(2r-1)$ stencil. The global smoothness measure τ_5, devised as the absolute difference between β_0 and β_2, is $\mathcal{O}(\Delta x^5)$ and

$$\frac{\tau_5}{\beta_k + \varepsilon} = \mathcal{O}(\Delta x^3),\ k = 0, 1, 2. \tag{15}$$

The condition $w_k - d_k = \mathcal{O}(\Delta x^3)$ is satisfied provided there are no critical points. Despite the limited success in enhancing accuracy near critical points, it was demonstrated that the new weight converges much faster to the optimal weight than classical WENO-JS in smooth regions, and that the new WENO-Z scheme performs better than WENO-JS and WENO-M. Furthermore, it was suggested to set $q = 1$ in Eq. (14) as a larger value of q results in a too dissipative scheme in non-smooth regions [20].

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2.3. Sixth-order WENO-CU6 and WENO-CU6-M schemes

In order to ameliorate the built-in dissipation from the background linear scheme, Hu and Adams [23] propose to admit the contribution from the downwind stencil, constructing a sixth-order WENO-CU6 scheme which adapts between central and upwind schemes smoothly by a new weighting strategy. Following the previous discussion, the approximate numerical flux at the cell face using the downwind candidate stencil gives

$$\hat{f}_{3,i+1/2} = \frac{1}{6}(11f_{i+1} - 7f_{i+2} + 2f_{i+3}). \tag{16}$$

Inspired by Borges et al. [20], the nonlinear weights are computed by

$$w_k = \frac{\alpha_k}{\sum_{k=0}^{3}\alpha_k},\ \alpha_k = d_k\left(C + \frac{\tau_6}{\beta_k + \varepsilon}\right)^q,\ k = 0, 1, 2, 3, \tag{17}$$

where $C = 20$ and $q = 1$ are parameters, and the optimal weights $d_0 = \frac{1}{20}$, $d_1 = \frac{9}{20}$, $d_2 = \frac{9}{20}$, $d_3 = \frac{1}{20}$ guarantee that the combined scheme recovers to the standard sixth-order central scheme with 6-point stencil. The smoothness indicator β_3 is evaluated from the full 6-point stencil as

$$\begin{aligned}
\beta_3 = \beta_6 &= \frac{1}{120960}\big[271779f_{i-2}^2 + f_{i-2}(-2380800f_{i-1} + 4086352f_i - 3462252f_{i+1} \\
&+ 1458762f_{i+2} - 245620f_{i+3}) + f_{i-1}(5653317f_{i-1} - 20427884f_i \\
&+ 17905032f_{i+1} - 7727988f_{i+2} + 1325006f_{i+3}) \\
&+ f_i(19510972f_i - 35817664f_{i+1} + 15929912f_{i+2} - 2792660f_{i+3}) \\
&+ f_{i+1}(17195652f_{i+1} - 15880404f_{i+2} + 2863984f_{i+3}) \\
&+ f_{i+2}(3824847f_{i+2} - 1429976f_{i+3}) + 139633f_{i+3}^2\big]
\end{aligned} \tag{18}$$

rather than from the generic formula Eq. (12) with 3-point stencil. The basic idea is that the contribution of the downwind stencil should vanish as long

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as the full 6-point stencil contains a discontinuity. The new global reference smoothness indicator τ_6 is defined as

$$\tau_6 = \beta_6 - \frac{1}{6}(\beta_0 + \beta_2 + 4\beta_1), \tag{19}$$

where

$$\tau_6 = \mathcal{O}(\Delta x^6),\ \frac{\tau_6}{\beta_k + \varepsilon} = \mathcal{O}(\Delta x^4),\ k = 0, 1, 2, 3. \tag{20}$$

As $w_k - d_k = \mathcal{O}(\Delta x^4)$, the requirement for maintaining overall sixth-order accuracy is satisfied. Hu and Adams [24] further propose another modified version WENO-CU6-M with $C = 1000$ and $q = 4$ by incorporating a physically-motivated scale-separation mechanism. This larger integer power of the smoothness-measure serves to separate the marginally-resolved and non-resolved flow scales. Numerical results imply that the new scheme functions as a physically-consistent implicit SGS model while retaining the shock-capturing capability.

3. Construction of targeted ENO scheme

In this section, we introduce a family of targeted ENO schemes. The general procedure to construct arbitrarily high-order schemes is discussed first; we then design five-point and six-point stencil targeted ENO schemes in detail.

3.1. Systematic formulation of targeted ENO schemes

3.1.1. Candidate stencils with incremental width

In analogy with the classical WENO scheme, a set of candidate stencils has to be derived for the high-order reconstruction. A new approach, by

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which the high-order full stencil is constructed from low-order stencils with incrementally increasing width, is proposed as shown in Fig. 1. In addi-

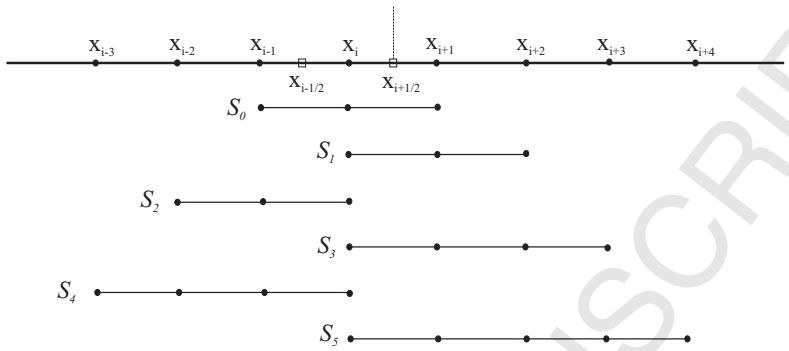


Figure 1: Candidate stencils with incremental width towards high-order reconstruction.

tion to the first three-point stencil S_0, each candidate stencil with one end node at x_i contains at least one-point upwind, thus decreasing the strong anti-dissipation induced by a pure downwind stencil. **In Fig. 2, we compare sketches for the reconstruction of 7th-order schemes for the classical WENO scheme and the proposed TENO scheme in the case of closely located two shock waves. For the classical WENO scheme all candidate stencils are crossed by the two discontinuities. The weighting strategy of the classical WENO schemes fails to guarantee the ENO property. However, for the TENO scheme the three-point candidate stencil S_0 avoids the crossing of discontinuities and thus can be singled out as smooth candidate stencil to contribute to the final reconstruction. Since practical numerical experiments reveal that fifth-order WENO scheme is robust for such configurations, the TENO schemes also can recover the robustness of fifth-order WENO scheme even for very-high-order**

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reconstructions.

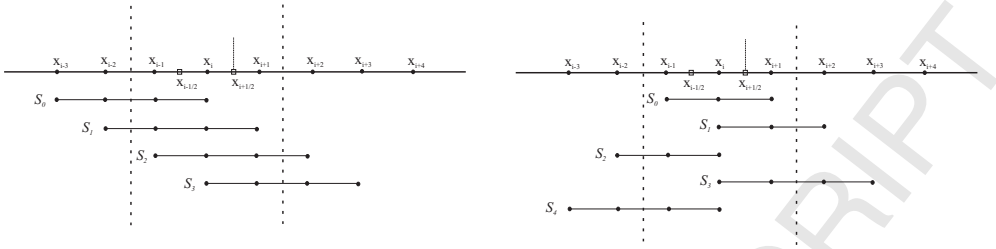


Figure 2: **Sketch of reconstruction schemes with closely located shock waves: the dashed lines represent two shock discontinuities. Classical 7th-order WENO scheme (left) and 7th-order TENO scheme (right).**

We redefine the symbol r as the width of a specific candidate stencil. Conservative numerical fluxes $\widehat{f}_{k,i+1/2}$ on different candidate stencils are evaluated with Eq. (7) in the same way as for WENO. The repective coefficients are presented in Table 1 up to $k = 5$.

The sequence of stencil widths r of low-order candidate stencils k for maximum order $K \geq 3$ on the full stencil is given as

$$\{r_k\} = \begin{cases} \{\underbrace{3, 3, 3, 4, \cdots, \frac{K+2}{2}}_{k=0,\cdots,K-3}\}, & \text{if } \mod(K, 2) = 0, \\ \{\underbrace{3, 3, 3, 4, \cdots, \frac{K+1}{2}}_{k=0,\cdots,K-3}\}, & \text{if } \mod(K, 2) = 1. \end{cases} \tag{21}$$

In the context of a K-point full stencil, an order K reconstruction can be achieved by the stencil combination $\bigcup_{k=0}^{K-3} S_k$. A $(K+1)$-order scheme is reconstructed from the stencil combination $\bigcup_{k=0}^{K-3} S_k$ with one additional upwind or downwind-biased stencil S_{K-2}. Arbitrarily high-order odd (upwind) or even (central) schemes can be devised by this recursion. Optimal weights

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Table 1: Numerical flux $\widehat{f}_{k,i+1/2} = \sum_m c_m f_m$ evaluated from the candidate stencil S_k

$\widehat{f}_{k,i+1/2}$	c_{i-3}	c_{i-2}	c_{i-1}	c_i	c_{i+1}	c_{i+2}	c_{i+3}	c_{i+4}
$k=0$			$-\frac{1}{6}$	$\frac{5}{6}$	$\frac{2}{6}$			
$k=1$				$\frac{2}{6}$	$\frac{5}{6}$	$-\frac{1}{6}$		
$k=2$		$\frac{2}{6}$	$-\frac{7}{6}$	$\frac{11}{6}$				
$k=3$				$\frac{3}{12}$	$\frac{13}{12}$	$-\frac{5}{12}$	$\frac{1}{12}$	
$k=4$	$-\frac{3}{12}$	$\frac{13}{12}$	$-\frac{23}{12}$	$\frac{25}{12}$				
$k=5$				$\frac{12}{60}$	$\frac{77}{60}$	$-\frac{43}{60}$	$\frac{17}{60}$	$-\frac{3}{60}$

for maximum order up to $K = 8$ on the full-point stencil are given in Table 2.

3.1.2. Scale separation

Motivated by Hu et al. [24] and Borges et al. [20], the smoothness measure is defined as

$$\gamma_k = \left(C + \frac{\tau_K}{\beta_{k,r} + \varepsilon} \right)^q, \ k = 0, \cdots, K-3, \tag{22}$$

where τ_K is the global reference smoothness indicator. We set $C = 1$ and the integer power q is defined as a large value, typically $q = 6$. With both WENO-Z and WENO-CU6, $q > 1$ is avoided as the resulting strong adaptation introduces excessive nonlinear dissipation at intermediate and high wave numbers. WENO-CU6-M sets $q = 4$ to incorporate a physically-motivated

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Table 2: Optimal weights for the upwind or central scheme for maximum order K.

Order	d_0	d_1	d_2	d_3	d_4	d_5
$K=3$	1					
$K=4$	$\frac{3}{6}$	$\frac{3}{6}$				
$K=5$	$\frac{6}{10}$	$\frac{3}{10}$	$\frac{1}{10}$			
$K=6$	$\frac{9}{20}$	$\frac{6}{20}$	$\frac{1}{20}$	$\frac{4}{20}$		
$K=7$	$\frac{18}{35}$	$\frac{9}{35}$	$\frac{3}{35}$	$\frac{4}{35}$	$\frac{1}{35}$	
$K=8$	$\frac{30}{70}$	$\frac{18}{70}$	$\frac{4}{70}$	$\frac{12}{70}$	$\frac{1}{70}$	$\frac{5}{70}$

* The d_k represents the optimal weight for candidate stencil S_k to assemble a K-th order scheme.

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scale-separation mechanism while setting $C = 1000$ for compensation. Such a choice reduces the effectiveness of discontinuity detection, which in turn compromises the robustness in simulations with strong shock waves. On the contrary, larger integer power exponent q and smaller C are preferable for an even stronger separation between resolved and non-resolved scales in the present study. The discontinuity-detection capability is significantly enhanced rather than sacrificed. Low dissipation for smooth regions is maintained by a tailored ENO-like stencil selection in the next section.

$\beta_{k,r}$ can be evaluated following Jiang and Shu [9] as

$$\beta_{k,r} = \sum_{j=1}^{r-1} \Delta x^{2j-1} \int_{x_{i-1/2}}^{x_{i+1/2}} \left(\frac{d^j}{dx^j} \hat{f}_k(x) \right)^2 dx. \tag{23}$$

Taylor expansion of $\beta_{k,r}$ gives[17]

$$\beta_{k,r} = \sum_{\ell=1}^{\lfloor \frac{r+1}{2} \rfloor} Q_{\beta_{2\ell}}(x_i) \Delta x^{2\ell} + B_{\beta_{k,r}} \left(\frac{df}{dx} \frac{d^r f}{dx^r} \right) \Big|_{x=x_i} \Delta x^{r+1} + \mathcal{O}(\Delta x^{r+2}), \ r = 3, \cdots, \tag{24}$$

where the first term on the right hand represents the common part for all stencils and $\lfloor \rfloor$ is the Gauss bracket. Provided that the flow is sufficiently smooth, the leading term of $\beta_{k,r}$ is $\mathcal{O}(\Delta x^2)$ for candidate stencils with different widths. In non-smooth regions, candidate stencils with larger width r are more likely to be crossed by discontinuities with $\beta_{k,r}$ of $\mathcal{O}(1)$. In this case the scheme is driven towards low-order reconstruction with smaller stencil-width. Since the first three stencils are the same as that of the classical fifth-order WENO-JS, the desirable robustness of WENO-JS is recovered in this case.

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3.1.3. ENO-like stencil selection

The smoothness measure is normalized first as

$$\chi_k = \frac{\gamma_k}{\sum_{k=0}^{K-3} \gamma_k}, \tag{25}$$

and then subjected to a sharp cutoff function

$$\delta_k = \begin{cases} 0, & \text{if } \chi_k < C_T, \\ 1, & \text{otherwise.} \end{cases} \tag{26}$$

The cutoff ensures that a candidate stencil is fully suppressed for the final reconstruction if the measured smoothness is below a certain threshold, otherwise it is adopted with its original weight. The choice of the threshold C_T can be determined from a compromise between good spectral properties and numerical robustness for discontinuity detection through quasi-linear spectral analyses and numerical experiments respectively, and will be discussed in the subsequent sections.

Note that parameter C_T also provides an effective and direct means to control the spectral properties of the underlying scheme for specific problems, e.g. compressible turbulence simulation with embedded shocklets, demanding for extremely low dissipation and slight nonlinear adaption.

3.1.4. Assembled high-order reconstruction scheme

The optimal weights d_k subjected to the cut-off δ_k are re-normalized as

$$w_k = \frac{d_k \delta_k}{\sum_{k=0}^{K-3} d_k \delta_k}, \tag{27}$$

so that the contributions from stencils containing discontinuities vanish. The final K-th order reconstructed numerical flux evaluated at cell face $i + \frac{1}{2}$ is

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assembled as

$$\hat{f}^K_{i+1/2} = \sum_{k=0}^{K-3} w_k \hat{f}_{k,i+1/2}. \tag{28}$$

Since the δ_k function switches between 0 and 1, a smooth-transition from high-order to low-order through the weighting strategy is prevented. Nevertheless, the resulting scheme can achieve arbitrarily high-order accuracy while satisfying the ENO property. For this reason, we call the new scheme "targeted ENO" (TENO) scheme.

3.1.5. Formal order of accuracy in smooth regions

Due to the ENO-like stencil selection, a sufficient condition for the overall scheme to restore its formal order of accuracy in smooth regions is $\chi_k \geq C_T$, which can be satisfied by the condition that all γ_k converges to C of Eq. (22) when Δx approaches zero. It is equivalent to the condition

$$\frac{\tau_K}{\beta_{k,r} + \varepsilon} = O(\Delta x^s), \quad s > 0. \tag{29}$$

This is a much weaker condition than those required for previous WENO schemes [19][20][24]. Note that such condition leads to relative flexible choice of the global reference smoothness indicator τ_K, as given in the following subsections and Appendix A, which ameliorates the order-degeneration problem suffered by classical WENO schemes.

3.2. Fifth-order TENO scheme

For a five-point TENO scheme, the involved candidate stencils $\{S_0, S_1, S_2\}$ are the same as for the classical fifth-order WENO-JS scheme. $\{\beta_{0,3}, \beta_{1,3}, \beta_{2,3}\}$ are calculated from Eq. (13), and the fifth-order undivided difference τ_5 is

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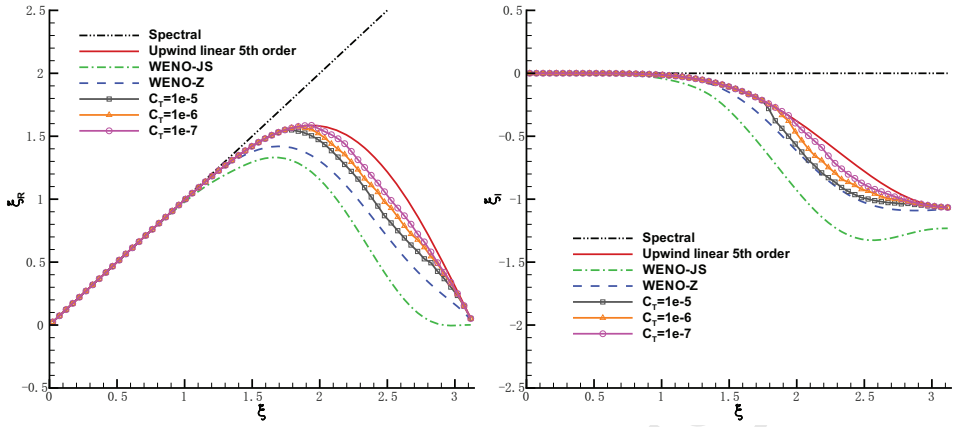


Figure 3: Approximate dispersion and dissipation properties for five-point stencil schemes: dispersion (left) and dissipation (right).

taken from Eq. (14) in WENO-Z. It is easy to find that τ_5 satisfies the condition Eq. (29) for the formal order of accuracy in smooth regions even near critical points.

The spectral properties of various five-point stencil schemes are shown in Fig. 3, following the ADR analysis [16]. The fifth-order TENO schemes with different thresholds constant C_T have much better spectral resolution than the WENO-JS and the WENO-Z schemes for almost the entire wave-number range. The dispersion and dissipation property of the fifth-order linear scheme is recovered up to wave numbers 1.75, 1.85 and 1.95, with $C_T = 10^{-5}$, 10^{-6} and 10^{-7} respectively. The wave-resolution capability of the TENO scheme follows that of the background linear scheme for low to intermediate wave numbers, while nonlinear dissipation is restricted to the high wave-number region.

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3.3. Sixth-order TENO scheme

Considering the six-point stencil TENO scheme, the candidate stencils $\{S_0, S_1, S_2, S_3\}$ with the stencil width r distribution $\{3, 3, 3, 4\}$ are quite different from that of the WENO-CU6 scheme with $\{3, 3, 3, 3\}$. Note that, unlike the smoothness indicator β_3 of the downwind stencil in WENO-CU6, $\{\beta_{0,3}, \beta_{1,3}, \beta_{2,3}, \beta_{3,4}\}$ is evaluated from Eq. (22) without exception. This simplification benefits from the new reconstruction approach, where candidate stencils with incremental width have at least one-point upwind. In this paper, Eq. (19) from WENO-CU6 is employed as sixth-order undivided difference τ_6. It is easy to find that τ_6 satisfies the condition Eq. (29) for the formal order of accuracy in smooth regions even near critical points.

Fig. 4 describes the spectral properties of various six-point stencil schemes. In comparison with the WENO-CU6 scheme, which departs from the linear scheme in both dispersion and dissipation at wave-number 1.2, the sixth-order TENO schemes delay the transition wave-number significantly. The TENO6 scheme with cutoff threshold C_T of 10^{-6} has the largest dissipation in high wave-number region, followed by WENO-CU6. WENO-CU6-M exhibits the lowest dispersion and dissipation among all the underlying six-point schemes.

4. Optimization of the TENO schemes

In this section, a generally effective optimization procedure of wave-resolution properties for both linear and nonlinear schemes is discussed. It is applied to the five- and six-point TENO schemes proposed in the previous section.

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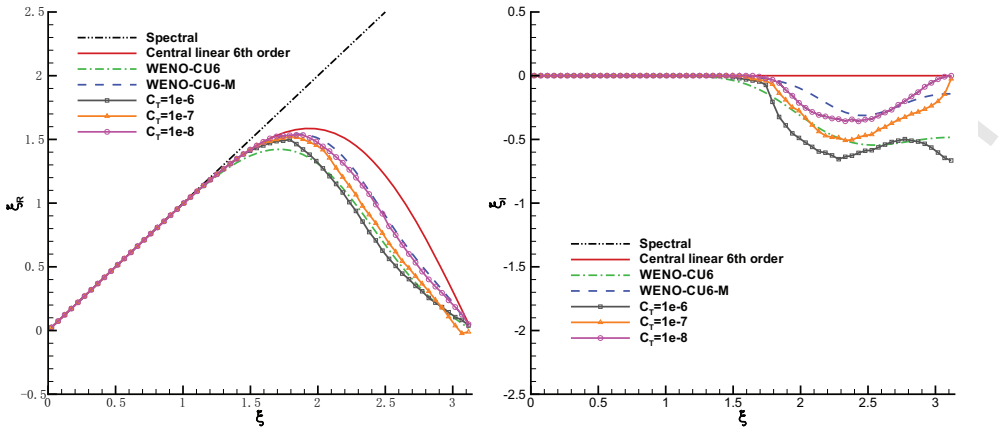


Figure 4: Approximate dispersion and dissipation properties for six-point stencil schemes: dispersion (left) and dissipation (right).

4.1. Guideline for optimizing linear and nonlinear schemes

Dissipation of the background linear scheme is a main dissipation source for the nonlinear shock-capturing scheme, so that low-dissipation and low-dispersion schemes always are preferable [26]. With an effective optimization strategy the generally-used traditional finite difference schemes can be optimized with respect to their dispersion and dissipation properties. Weirs and Candler [28] construct a six-point scheme with low dissipation and third-order accuracy by introducing the downwind stencil (WENO-SYMBO). Arshed and Hoffmann [22] propose the WENO-BO scheme by optimizing the weighted integral error for dispersion and dissipation within the intermediate wave-number region. However, both schemes produce anti-dissipation in the low to intermediate wave-number range, see Fig.1 of [22].

Following Pirozzoli's ADR analysis [16], Hu et al. [25] propose a dispersion-dissipation relation for a general finite difference scheme to estimate the

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numerical dissipation necessary to damp spurious high wave-number errors from

$$\zeta = \frac{\left|\frac{d\xi_R}{d\xi} - 1\right|}{-\xi_I} \approx \mathcal{O}(10) \tag{30}$$

where ξ_R and ξ_I is the real and image part of the modified wave number for the non-resolved wave-number range. Although this relation is obtained from a linear analysis, application to nonlinear schemes is fairly straightforward. Thus it becomes a guideline for optimizing linear and nonlinear schemes to possess appropriate dissipation.

The dispersion-dissipation relation of various nonlinear schemes is shown in Fig. 5. Due to the excessive dissipation of the fifth-order upwind linear scheme, $\zeta \leqslant 5.5$ for all the nonlinear five-point schemes, indicating that they are free from spurious waves. High wave-number Fourier modes supported by the utilized mesh resolution are rapidly damped, especially for the classical fifth-order WENO-JS, as ζ decreases rapidly in the high wave-number region. WENO-Z and TENO5 produce relatively smaller dissipation but still are too dissipative, e.g., for turbulence simulations. For the purpose of implicit large eddy simulations, as clarified by Zhao et al. [14], no notable advantages of the improved WENO-Z could be found compared with WENO-JS.

For the six-point stencil schemes, $\zeta \approx \mathcal{O}(100)$ between wave-number 1.1 and 1.8, as the background sixth-order central scheme is non-dissipative. For preventing the proliferation of numerical errors owing to nonlinear transport, it is crucial to properly balance the dissipation and dispersion by a reasonable choice of ζ, so that physically-consistent schemes could be obtained. An interesting observation is that the ζ distributions of both WENO-SYMBO and WENO-BO apparently fail to satisfy the dispersion-dissipation relation

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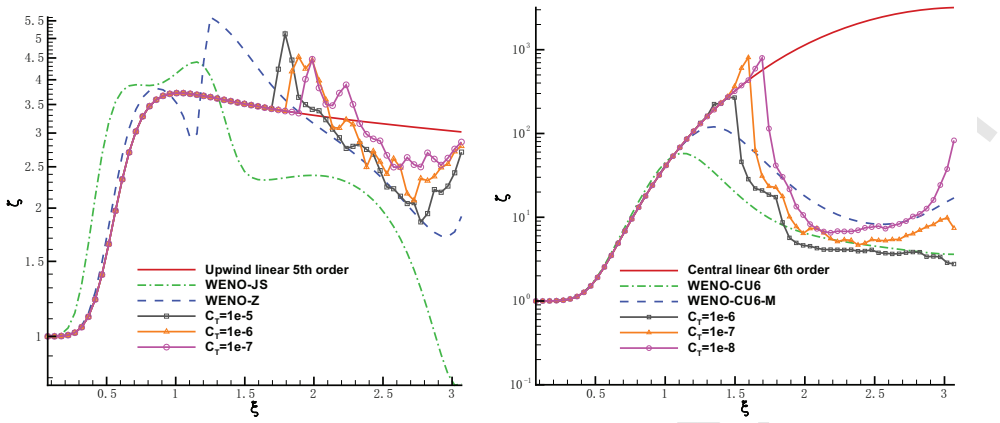


Figure 5: Approximate dispersion-dissipation relation for various nonlinear schemes: five-point stencil schemes (left) and six-point stencil schemes (right). y axis is in log form and $\zeta = \frac{\left|\frac{d\xi_R}{d\xi}-1\right|+10^{-3}}{-\xi_I+10^{-3}}$ approximately.

[29].

Here, we present a simple way to optimize the background scheme following an appropriate dispersion-dissipation relation. Instead of minimizing the dispersion errors, an order K scheme can be optimized by assembling two sub-schemes of order $(K-1)$ as

$$f^{opt}_{i+\frac{1}{2}} = \eta g^{K-1}_{i+\frac{1}{2}}(1) + (1-\eta) g^{K-1}_{i+\frac{1}{2}}(2). \tag{31}$$

For even K, the sub-schemes are upwind biased and downwind biased featuring the same dispersion errors. Since the two sub-schemes have the same dispersion error, the optimization procedure adjusts dissipation only to achieve the optimization objective. On the other hand, for odd K, the dispersion error also is affected as the sub-schemes possess different dispersion properties. The combination of these two schemes leads to lower dissipation than the order K standard scheme while introducing extra dispersion errors. Since

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ζ is the optimization objective function, such slightly increased dispersion errors are tolerated and do not pollute the solution according to [25].

4.2. Optimization of the fifth-order TENO scheme

Based on the five-point stencil, the two fourth-order sub-schemes are given as

$$\begin{aligned} g^4_{i+\frac{1}{2}}(1) &= \tfrac{1}{12}(f_{i-2} - 5f_{i-1} + 13f_i + 3f_{i+1}), \\ g^4_{i+\frac{1}{2}}(2) &= \tfrac{1}{12}(-f_{i-1} + 7f_i + 7f_{i+1} - f_{i+2}), \end{aligned} \tag{32}$$

where $g^4_{i+\frac{1}{2}}(1)$ is an upwind scheme and $g^4_{i+\frac{1}{2}}(2)$ is a central non-dissipative scheme. The combined overall scheme recovers the classical upwind fifth-order scheme when $\eta = 0.4$. Gradually decreasing η drives the overall scheme biased towards a central scheme as shown the Table 3, which gives the corresponding optimal weights for incremental-width stencils. The spectral property of these optimized schemes is shown in Fig. 6. As η decreases from 0.35 to 0.2, the dissipation property is greatly improved, and the extremum of ζ increases from approximately 5.5 to 16.5, maintaining order $\mathcal{O}(10)$.

Table 3: Optimal weights based on the dispersion-dissipation condition for five-point-stencil schemes

η	d_0	d_1	d_2
0.35	0.5875	0.325	0.0875
0.30	0.5750	0.350	0.0750
0.25	0.5625	0.375	0.0625
0.20	0.5500	0.400	0.0500

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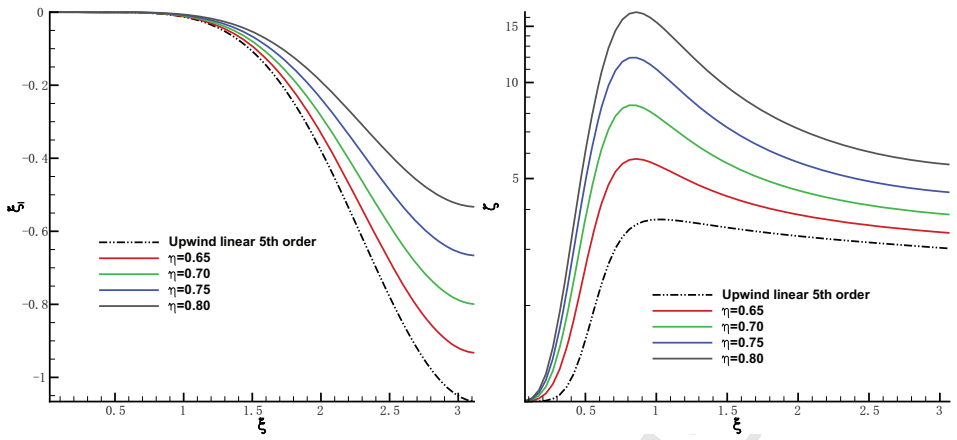


Figure 6: Spectral properties of the optimized linear five-point stencil schemes: dissipation (left) and dispersion-dissipation relation (right).

A straightforward implementation of optimal weights with $\eta = 0.2$ to the nonlinear TENO scheme proposed in previous section results in a nonlinear five-point optimized TENO scheme. As illustrated in Fig. 7, the nonlinear procedure induces extra dissipation only at high wave numbers and further decreases ζ in this region. The overall nonlinear TENO5 scheme is less dissipative than a classical fifth-order linear scheme especially for intermediate wave numbers, thus it is expected to resolve significantly more flow scales.

4.3. Optimization of the sixth-order TENO scheme

The two fifth-order sub-schemes for a six-point stencil are given as

$$\begin{aligned} g^5_{i+\frac{1}{2}}(1) &= \tfrac{1}{60}(2f_{i-2} - 13f_{i-1} + 47f_i + 27f_{i+1} - 3f_{i+2}), \\ g^5_{i+\frac{1}{2}}(2) &= \tfrac{1}{60}(-3f_{i-1} + 27f_i + 47f_{i+1} - 13f_{i+2} + 2f_{i+3}). \end{aligned} \tag{33}$$

As $g^5_{i+\frac{1}{2}}(1)$ and $g^5_{i+\frac{1}{2}}(2)$ possess the same dispersion property as the standard sixth-order central scheme the combined overall scheme maintains the dis-

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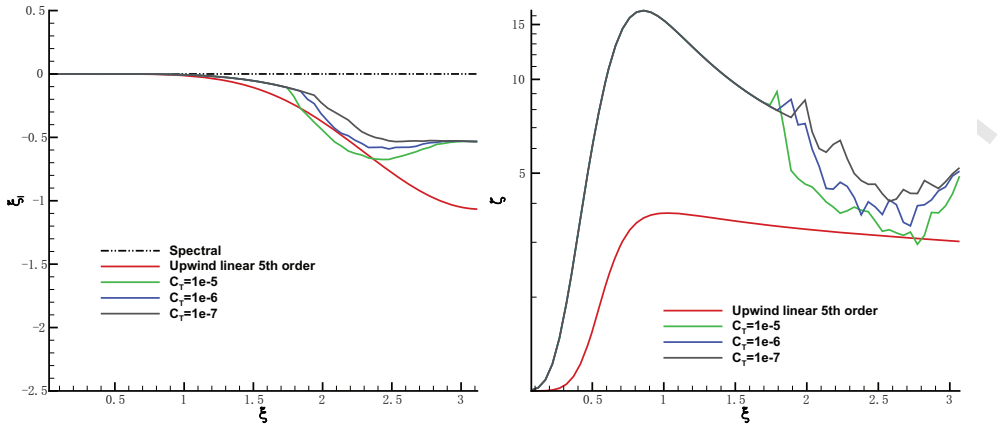


Figure 7: Spectral property of optimized nonlinear five-point stencil schemes: dissipation property (left) and dispersion-dissipation relationship (right).

persion errors. Compared to $\eta = 0.5$ which leads to a non-dissipative central scheme, the overall scheme exhibits correspondingly increasing dissipation with increasing η. Table 4 addresses the optimal weights for increasing stencil widths with specific η. Figure 8 shows the optimized spectral properties. Note that ζ exhibits a nonlinear growth tendency with decreasing η.

Rather than as shown in [25] the optimized fifth-order linear scheme is obtained by enforcing the constraint $\zeta = 10$ at $\xi = 1.302$ according to Eq. (30). Full-stencil scheme is constructed with $\eta = 0.54$ implying less dissipation. The spectral properties of the optimized nonlinear six-point TENO schemes is shown in Fig. 9. Throughout the entire wave-number range $\zeta \leq 35$, and decreases to less than 10 at high wave numbers.

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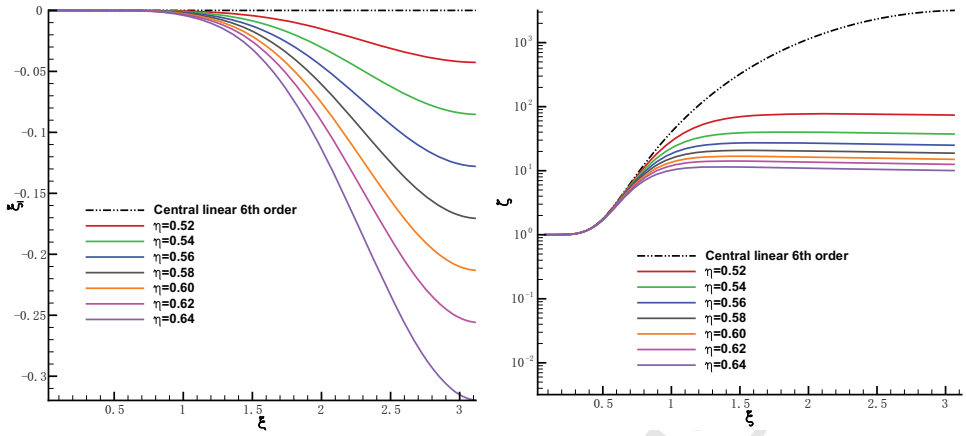


Figure 8: Spectral properties of optimized linear six-point stencil schemes: dissipation property (left) and dispersion-dissipation relation (right).

Table 4: Optimal weights based on the dispersion-dissipation condition for six-point-stencil schemes.

η	d_0	d_1	d_2	d_3
0.52	0.456	0.3	0.052	0.192
0.54	0.462	0.3	0.054	0.184
0.56	0.468	0.3	0.056	0.176
0.58	0.474	0.3	0.058	0.168
0.60	0.480	0.3	0.060	0.160
0.62	0.486	0.3	0.062	0.152
0.64	0.492	0.3	0.064	0.144

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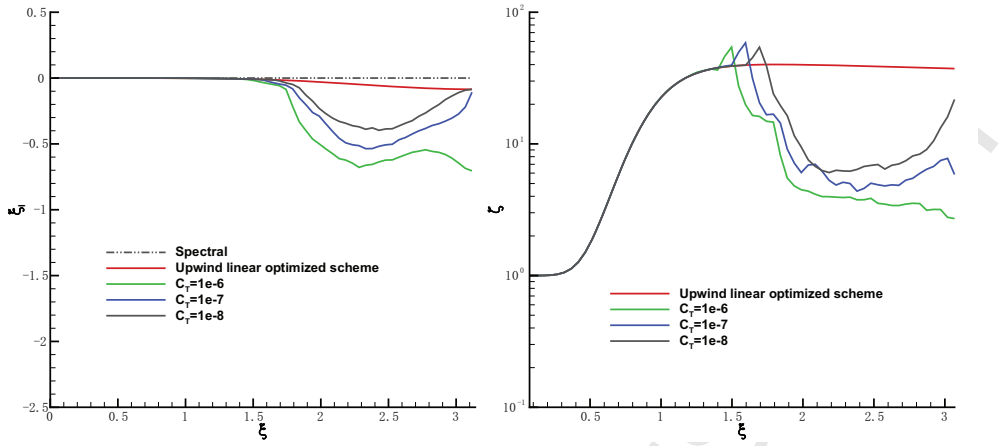


Figure 9: Spectral properties of the optimized nonlinear six-point stencil schemes: dissipation (left) and dispersion-dissipation relation (right).

5. Numerical validation

A variety of benchmark problems is considered to assess the proposed schemes. Compressible Navier-Stokes equations and Euler equations are applied as physical models for viscous and inviscid flow simulations. The ideal-gas equation of state $p = (\gamma - 1)\rho e$ with $\gamma = 1.4$ is adopted to close the conservation laws. As a default setting, the Roe average is exploited for characteristic decomposition at the cell face, and the Global Lax-Friedrichs(GLF) is utilized for flux splitting. For one-dimensional cases uniformly distributed grid points are used and the 3rd-order strongly stable Runge-Kutta [34] scheme is chosen for time integration. For two-dimensional cases the wavelet-based multi-resolution method [35] is applied with 2nd-order strongly stable Runge-Kutta scheme [34]. The fifth-order and sixth-order TENO schemes are compared with the well-validated WENO-JS, WEMO-Z, WENO-CU6 and WENO-CU6-M comprehensively. In all computations, a single set of

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parameters as given in Table 5 and a CFL number of 0.6 are applied .

Table 5: Parameters of TENO-family schemes for numerical validation

Scheme	d_0	d_1	d_2	d_3	C	q	C_T
TENO5	0.600	0.300	0.100	/	1.0	6	10^{-5}
TENO5-opt	0.550	0.400	0.050	/	1.0	6	10^{-5}
TENO6	0.450	0.300	0.050	0.200	1.0	6	10^{-7}
TENO6-opt	0.462	0.300	0.054	0.184	1.0	6	10^{-7}

5.1. ***Linear advection problem***

We first consider the one-dimensional Gaussian pulse advection problem [36] to validate the numerical order of proposed schemes. The linear advection equation

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0, \tag{34}$$

with initial condition

$$u_0(x) = e^{-300(x-x_c)^2}, \ x_c = 0.5, \tag{35}$$

is solved on a computational domain $0 \leq x \leq 1$.

Periodic boundary conditions are imposed at $x = 0$ and $x = 1$. The solution is advanced to $t = 1$ corresponding to one period in time, and a sequence of globally refined uniform grids is employed to investigate the L_∞-error convergence. The timestep

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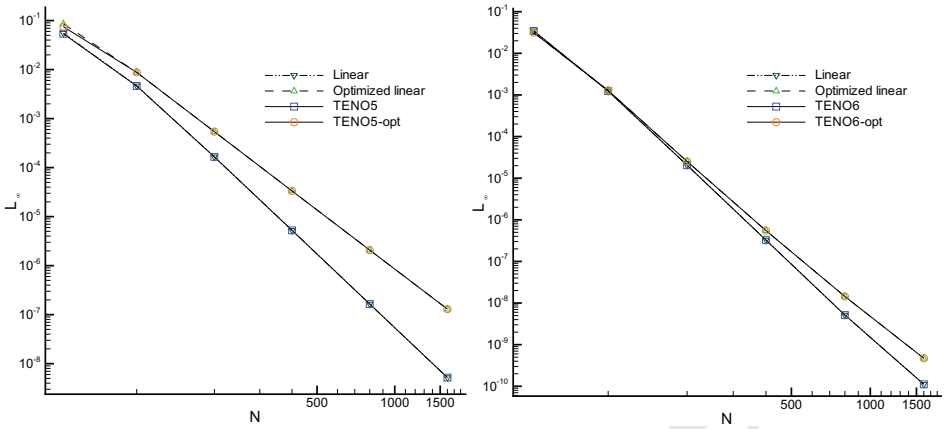


Figure 10: **Convergence of the L_∞-error: five-point stencil schemes (left) and six-point stencil schemes. The linear scheme represents the maximum-order background scheme of TENO and the optimized linear scheme represents the spectrally optimized background scheme of optimized TENO.**

is decreased so that the time-integration error can be neglected. As shown in Fig. 10, the L_∞-error convergence histories of all proposed TENO schemes coincide with their corresponding background linear schemes and therefore the desirable order of accuracy is achieved. Order-degeneration and instability of the classical WENO schemes, as shown in [36], are overcome. As expected, both the optimized linear and nonlinear schemes are by one order less accurate than the maximum-order accuracy standard schemes.

5.2. Shock-tube problem

Two typical shock-tube test problems, i.e. the Sod problem taken from [37] and the Lax problem from [38], are simulated to validate the shock-capturing capability of proposed schemes.

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The initial condition for the Sod problem is

$$(\rho, u, p) = \begin{cases} (1, 0, 1), & \text{if } 0 \le x < 0.5 \\ (0.125, 0, 0.1), & \text{if } 0.5 \le x \le 1 \end{cases}, \tag{36}$$

and the final simulation time is $t = 0.2$.

Fig. 11 shows the computed density distribution from five-point and six-point stencil schemes. The solutions from WENO-Z, TENO5 and TENO5-opt are obviously better than that of WENO-JS in resolving the contact discontinuity. The shock wave is captured stably and no overshoots or oscillations can be observed. Considering the six-point stencil schemes, slight oscillations appear near the contact discontinuity for WENO-CU6-M. TENO6 and TENO6-opt perform as well as WENO-CU6.

The initial condition for the Lax problem is

$$(\rho, u, p) = \begin{cases} (0.445, 0.689, 3.528), & \text{if } 0 \le x < 0.5 \\ (0.5, 0, 0.5710), & \text{if } 0.5 \le x \le 1 \end{cases}, \tag{37}$$

and the final simulation time is $t = 0.14$.

Fig. 12 gives a comparison of the computed density and velocity distribution. WENO-JS exhibits marginally larger dissipation than the others while WENO-CU6-M produces slight overshoots for the density profile near the contact discontinuity. All schemes preserve monotonicity near the shock wave.

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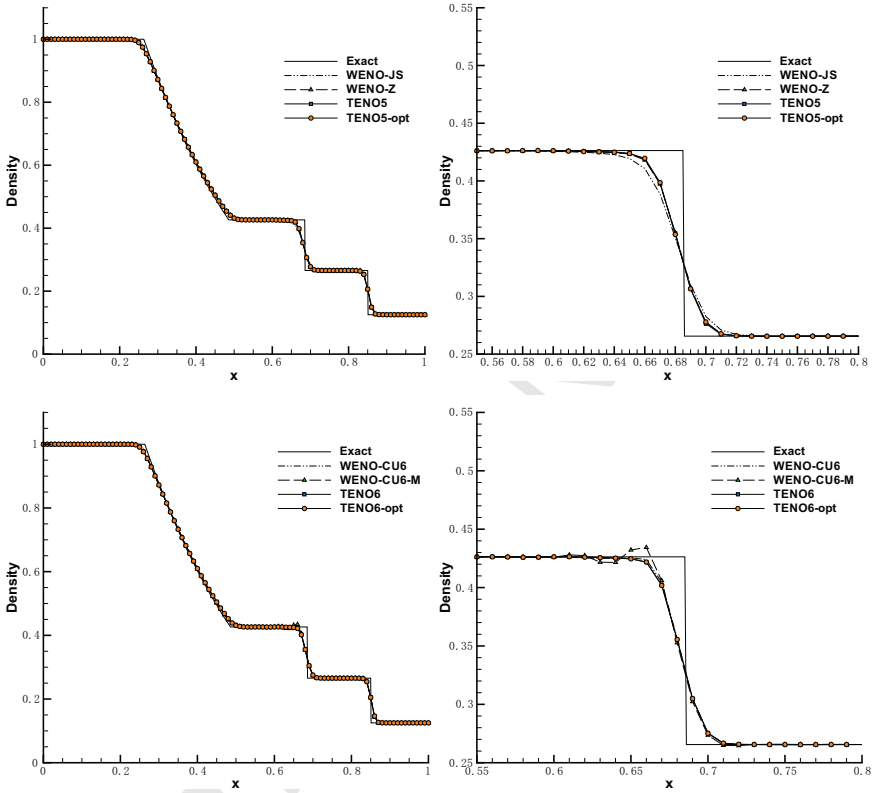


Figure 11: Sod shock-tube problem: density distribution (left) and a zoom of the region near contact discontinuity (right). Top: five-point stencil schemes. Bottom: six-point stencil schemes. Discretization on 100 uniformly distributed grid points. "Exact" represents the theoretical solution of the Riemann problem.

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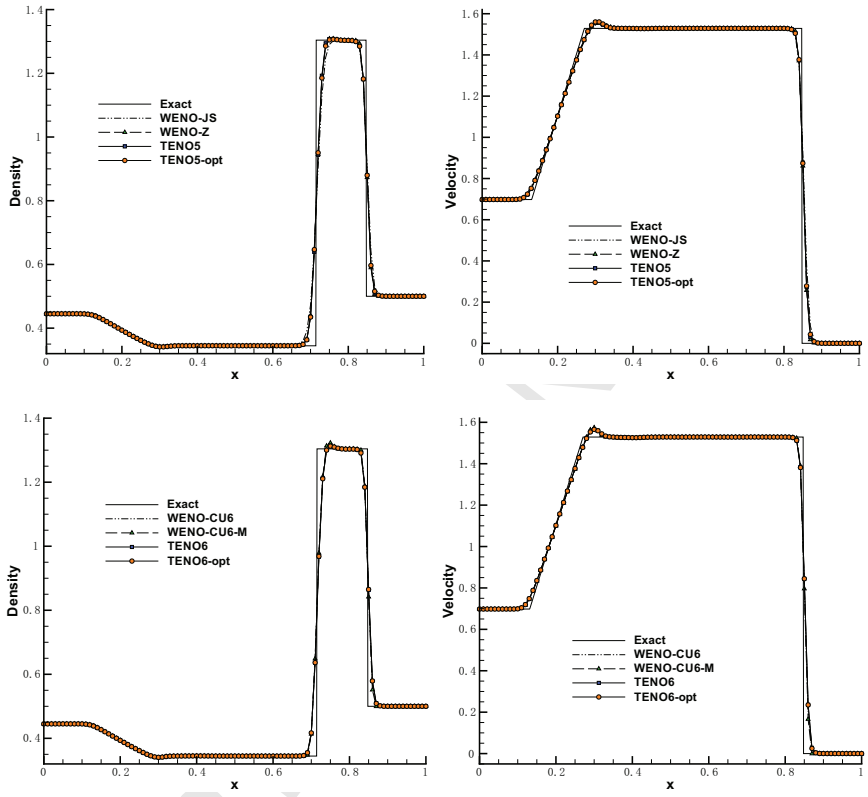


Figure 12: Lax shock-tube problem: density distribution (left) and velocity distribution (right). Top: five-point stencil schemes. Bottom: six-point stencil schemes. Discretization on 100 uniformly distributed grid points. "Exact" represents the theoretical solution of the Riemann problem.

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5.3. Interacting blast waves

The two-blast-wave interaction taken from Woodward and Colella [39] is considered. The initial condition is

$$(\rho, u, p) = \begin{cases} (1, 0, 1000), & \text{if } 0 \le x < 0.1, \\ (1, 0, 0.01), & \text{if } 0.1 \le x < 0.9, \\ (1, 0, 100), & \text{if } 0.9 \le x \le 1. \end{cases} \tag{38}$$

A reflective boundary condition is imposed at $x = 0$ and $x = 1$. The simulation is performed on a uniform mesh with $N = 400$ until final time $t = 0.038$. The reference "exact" solution is produced by the fifth-order WENO-JS scheme on a uniform mesh with $N = 2500$. A Roe scheme with entropy-fix is applied for evaluating the numerical flux.

In terms of five-point stencil schemes, TENO5 and TENO5-opt resolve the density profile much better than WENO-JS, especially near the valley $x = 0.75$ and the right peak $x = 0.78$ in Fig. 13. Furthermore, TENO5 and TENO5-opt exhibit less dissipation than WENO-Z, which can be seen at $x = 0.78$. For the pressure profile, TENO5 and TENO5-opt again provide much better results. With respect to the velocity profile, WENO-JS and WENO-Z produce strong overshoots near $x = 0.86$ while TENO5 and TENO5-opt perform well. Fig. 14 depicts the calculated results from six-point stencil schemes. Except that WENO-CU6-M fails to pass this case, the other three schemes provide an overall better result than that of five-point stencil schemes. WENO-CU6 produces spurious waves near $x = 0.75$ and $x = 0.86$ in the pressure profile, and overshoots at $x = 0.86$ in the velocity distribution. TENO6 and TENO6-opt however do not exhibit such spurious oscillations and resolve the density profile better at $x = 0.78$.

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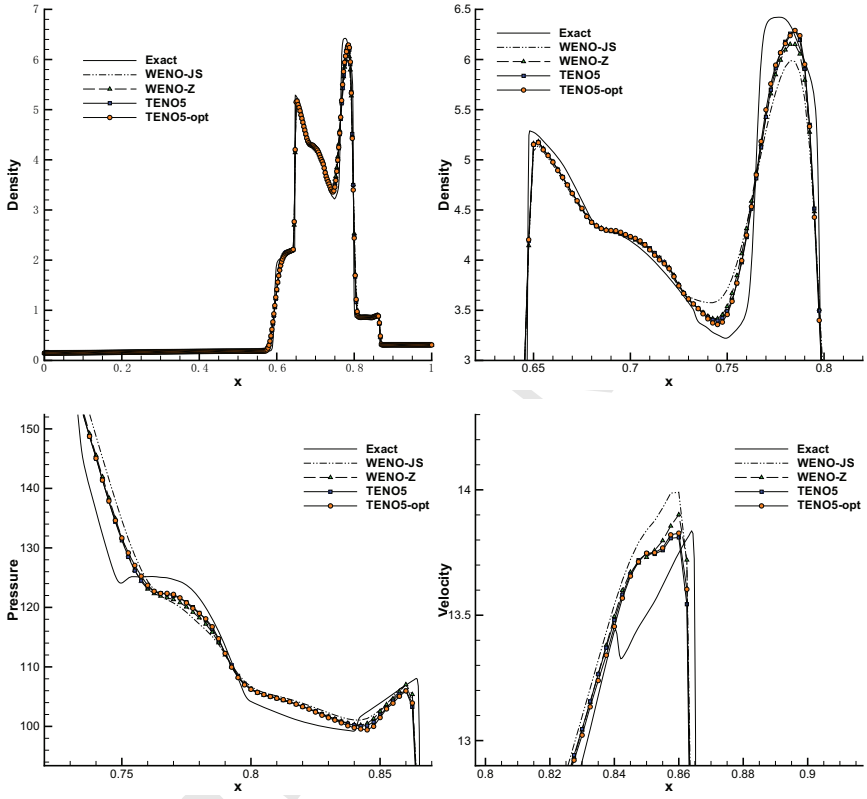


Figure 13: Interacting blast waves problem: solution from five-point stencil schemes. Top: density distribution (left) and a zoom of the density profile (right). Bottom: zoom of the pressure profile (left) and zoom of the velocity profile (right). Discretization on 400 uniform grid points.

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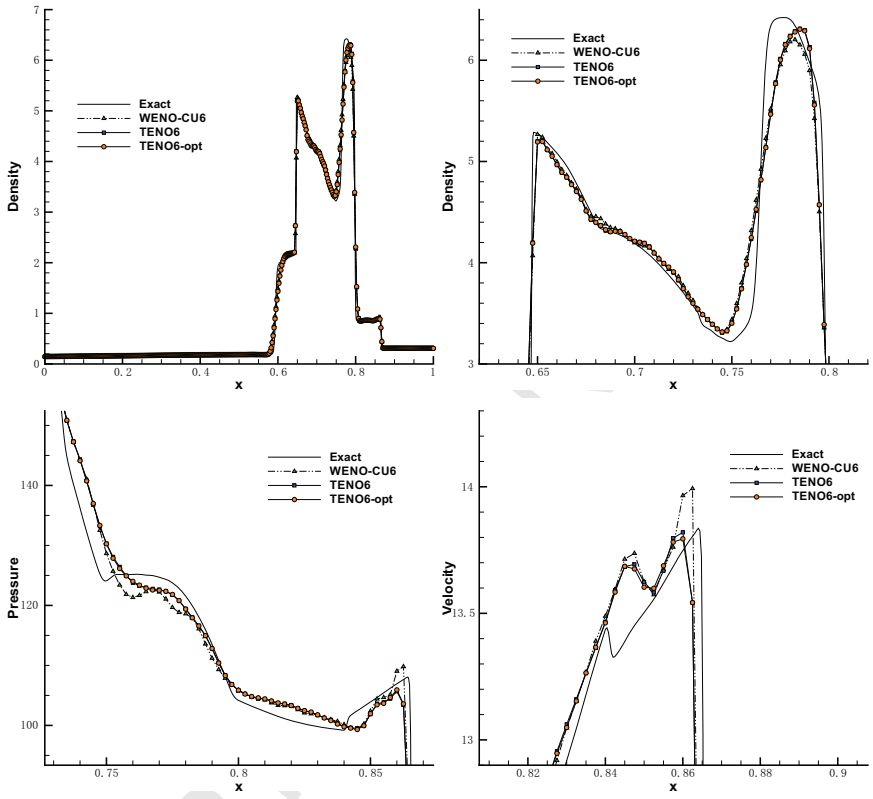


Figure 14: Interacting blast waves problem: solution from six-point stencil schemes. Top: density distribution (left) and a zoom of the density distribution (right). Bottom: zoom of the pressure profile (left) and zoom of the velocity profile (right). Discretization on 400 uniform grid points. WENO-CU6-M fails to pass this case.

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5.4. Shock-density wave interaction

This case is proposed by Shu and Osher [40]. A one-dimensional Mach 3 shock wave interacts with a perturbed density field generating both small-scale structures and discontinuities, hence it is selected to validate shock-capturing and wave-resolution capability. The initial condition is given as

$$(\rho, u, p) = \begin{cases} (3.857, 2.629, 10.333), & \text{if } 0 \leq x < 1, \\ (1 + 0.2\sin(5x), 0, 1), & \text{if } 1 \leq x \leq 10. \end{cases} \tag{39}$$

The computational domain is [0,10] with $N = 200$ uniformly distributed mesh cells and the final evolution time is $t = 1.8$. The reference "exact" solution is obtained by the fifth-order WENO-JS scheme with $N = 2000$.

As shown in Fig. 15, all schemes are capable of capturing the acoustic waves with shocklets successfully. Concerning the entropy waves, significant improvements can be observed from the new proposed schemes. Specifically, TENO5-opt shows the best resolution in maintaining the amplitudes of density waves, followed by TENO5, and then WENO-Z. The WENO-JS scheme smears the flow features substantially exhibiting the largest dissipation. For capturing the shock waves, TENO5 and TENO5-opt produce slightly better results than WENO-JS scheme as well. This implies that TENO5 and TENO5-opt feature much better wave-resolution properties and shock-capturing capability than the WENO-JS scheme. For six-point stencil schemes, numerical results further improve as indicated in Fig. 16. It can be noticed that TENO6, TENO6-opt and WENO-CU6-M are less dissipative than WENO-CU6. TENO6 obtains qualitatively almost the same result as WENO-CU6-M, retaining the entropy wave amplitude up to 4.52 which is higher than that of TENO5-opt at about 4.4. In terms of the shock region,

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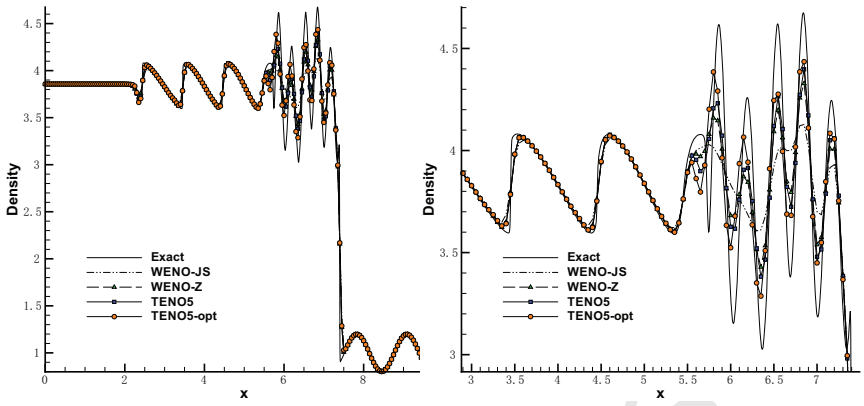


Figure 15: Shock-density wave interaction problem: solutions from five-point stencil schemes. Density distribution (left) and a zoom of the density distribution (right). Discretization on 200 uniformly distributed grid points.

WENO-CU6 and WENO-CU6-M suffer from slight numerical oscillations for both density and velocity profiles. TENO6 and TENO6-opt are free from such spurious waves while sharply capturing the shock waves.

5.5. Double Mach reflection of a strong shock

This two-dimensional case is taken from Woodward and Colella [39]. The initial condition is

$$(\rho, u, v, p) = \begin{cases} (1.4, 0, 0, 1), & \text{if } y < 1.732(x - 0.1667), \\ (8, 7.145, -4.125, 116.8333), & \text{otherwise.} \end{cases} \tag{40}$$

The computational domain is $[0, 4] \times [0, 1]$ and the final simulation time is $t = 0.2$. Initially, a right-moving Mach 10 shock wave is placed at $x = 0.1667$ with an incidence angle of 60^0 to the x-axis. The post-shock condition is imposed from $x = 0$ to $x = 0.1667$ whereas a reflecting wall condition is

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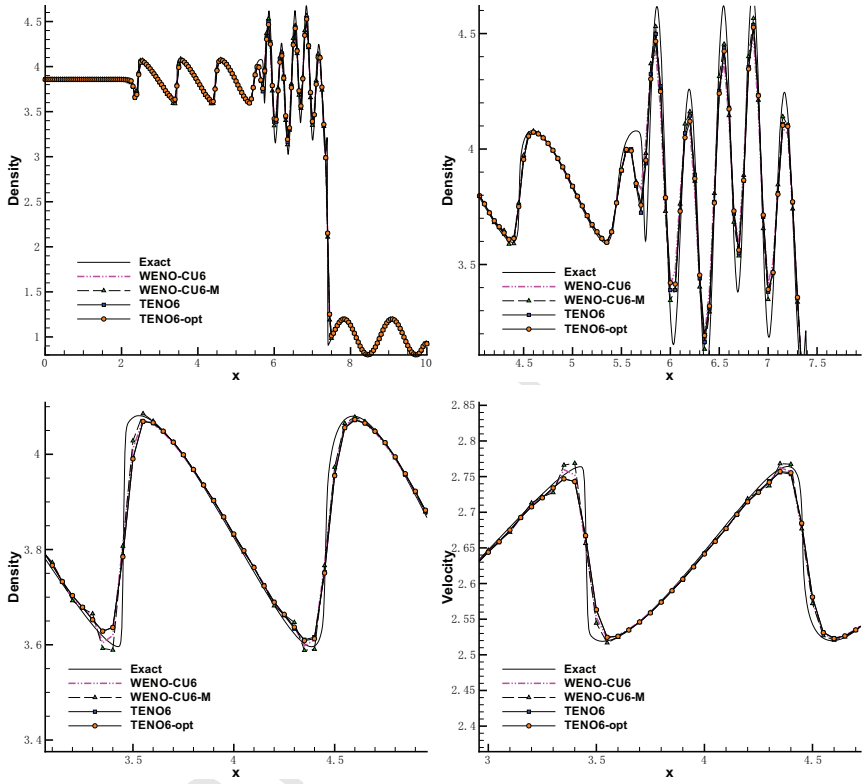


Figure 16: Shock-density wave interaction problem: solutions from six-point stencil schemes. Top: density distribution (left) and a zoom of the density distribution (right). Bottom: zoom of the density profile (left) and zoom of the velocity profile (right). Discretization on 200 uniformly distributed grid points.

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enforced from $x = 0.1667$ to $x = 4$ at the bottom. For the top boundary condition, the fluid variables are defined as to exactly follow the evolution of the Mach 10 shock wave. Generic inflow and outflow conditions are imposed for the left and right side of the computational domain. An adaptive multi-resolution mesh [35] is applied with the effective resolution of 1024×256, i.e. grid space $\Delta x = \Delta y = 1/256$.

As described in Fig. 17 and Fig. 18, only the domain region $[2, 3] \times [0, 0.9]$ is presented for the facilitation of comparison. Except that WENO-CU6-M fails the simulation, other schemes reproduce the solutions having significant differences in the region around the double Mach stems. Among all the five-point stencil schemes, TENO5-opt resolves the finest small-scale structures and the strongest wall jet, which are comparable to that from the fifth-order WENO-JS scheme at resolution $\Delta x = \Delta y = 1/960$, see Fig.1 in [41], followed by TENO5. When considering the results from TENO6, TENO6-opt and WENO-CU6, remarkable improvements in resolving small-scale structures can be observed in Fig. 18. Specifically, TENO6 produces the richest small-scale structures, comparable to that from the ninth-order WENO-JS scheme at resolution $\Delta x = \Delta y = 1/960$, see Fig.1 in [41]. TENO6-opt resolves slightly less structures than TENO6 due to the dissipation added in the optimization procedure. Nevertheless, the result is still much better than that of WENO-CU6.

5.6. Rayleigh-Taylor instability

As another canonical problem, which contains both discontinuities and complex flow structures, the invicid Rayleigh-Taylor instability case proposed

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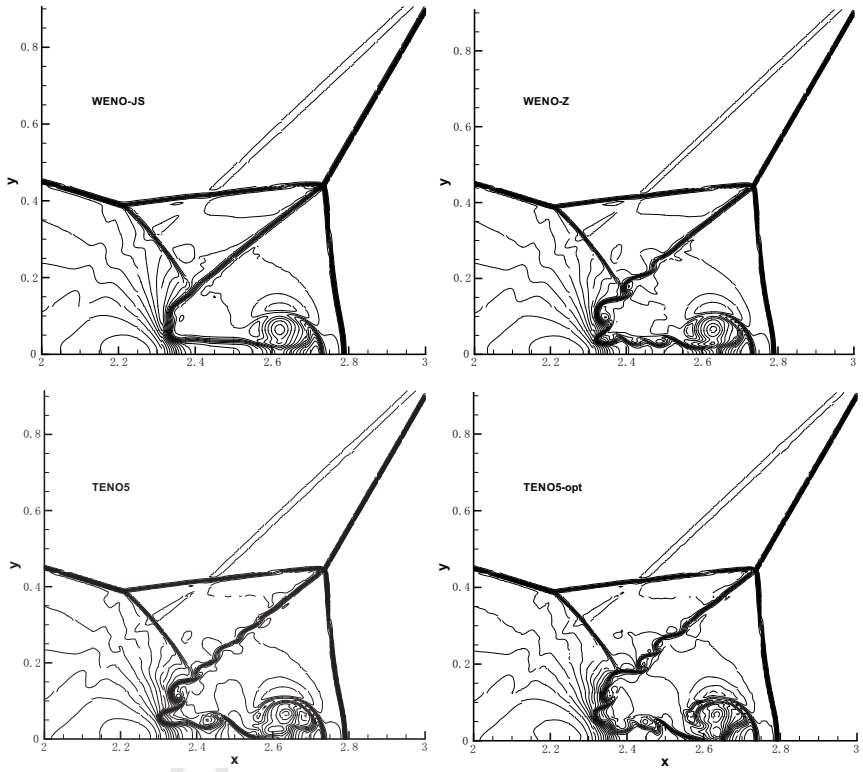


Figure 17: Double Mach reflection of a strong shock: density contours from five-point stencil schemes. Resolution 1024×256. This figure is drawn with 43 density contours between 1.887 and 20.9.

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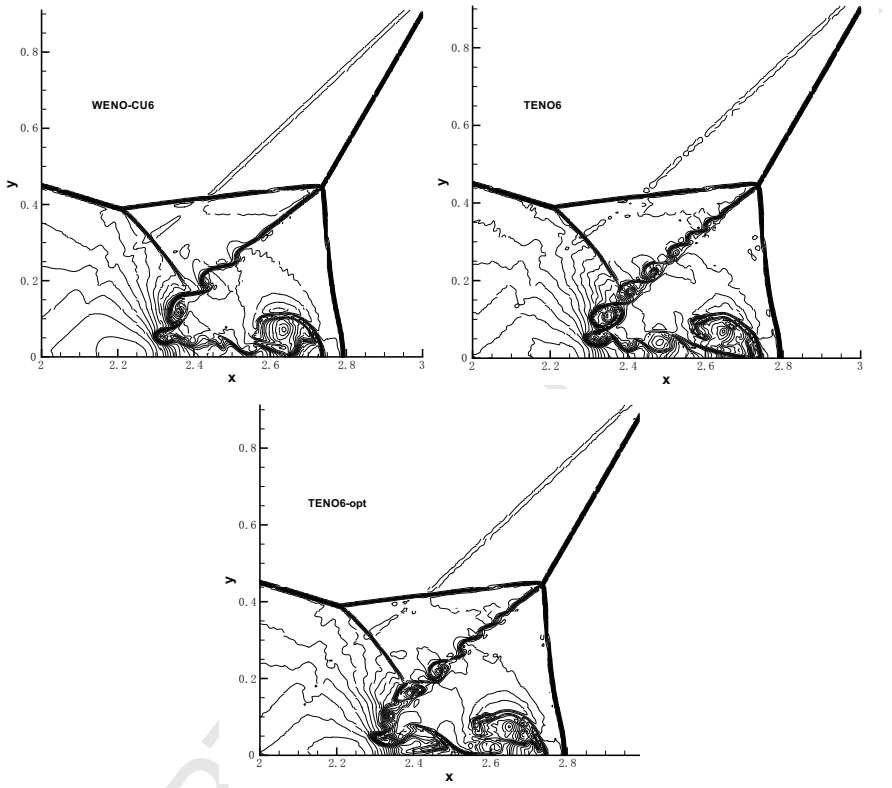


Figure 18: Double Mach reflection of a strong shock: density contours from six-point stencil schemes. Resolution 1024×256. This figure is drawn with 43 density contours between 1.887 and 20.9. WENO-CU6-M fails to pass this case.

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by Xu and Shu [42] is considered. The initial condition is

$$(\rho, u, v, p) = \begin{cases} (2, 0, -0.025c\cos(8\pi x), 1 + 2y), & \text{if } 0 \le y < 1/2, \\ (1, 0, -0.025c\cos(8\pi x), y + 3/2), & \text{if } 1/2 \le y \le 1, \end{cases} \tag{41}$$

where the sound speed is $c = \sqrt{\gamma \frac{p}{\rho}}$ with $\gamma = \frac{5}{3}$. The computational domain is $[0, 0.25] \times [0, 1]$. Reflective boundary conditions are imposed at the left and right side of the domain. Constant primitive variables $(\rho, u, v, p) = (2, 0, 0, 1)$ and $(\rho, u, v, p) = (1, 0, 0, 2.5)$ are set for the bottom and top boundaries, respectively. Initially, the interface located at y = 0.5 separates the heavy and light fluid, forming a contact discontinuity. For inviscid simulation solving Euler equations, the smaller magnitude of numerical dissipation from high-order shock-capturing scheme results in richer fine structures.

Fig. 19 shows the solutions from all 8 schemes at a resolution 128×512, i.e. grid space $\Delta x = \Delta y = 1/512$. TENO5-opt and TENO5 have resolved much richer vortical structures than WENO-JS and WENO-Z. Furthermore, the low-dissipation property of TENO5-opt and TENO5 induce symmetry breaking of flow field. A more dissipative scheme is more likely to preserve this symmetry even when increasing the resolution significantly, see Fig.2 in [41]. For six-point stencil schemes, further improved results are obtained especially with TENO6, TENO6-opt and WENO-CU6-M. When the mesh resolution is doubled in both coordinate directions, the solution from WENO-CU6 also loses symmetry in Fig. 20. Another noticeable outcome is that small structures resolved from TENO5-opt and TENO5 with resolution of 128×512 are comparable to that from WENO-JS and WENO-Z with resolution of 256×1024.

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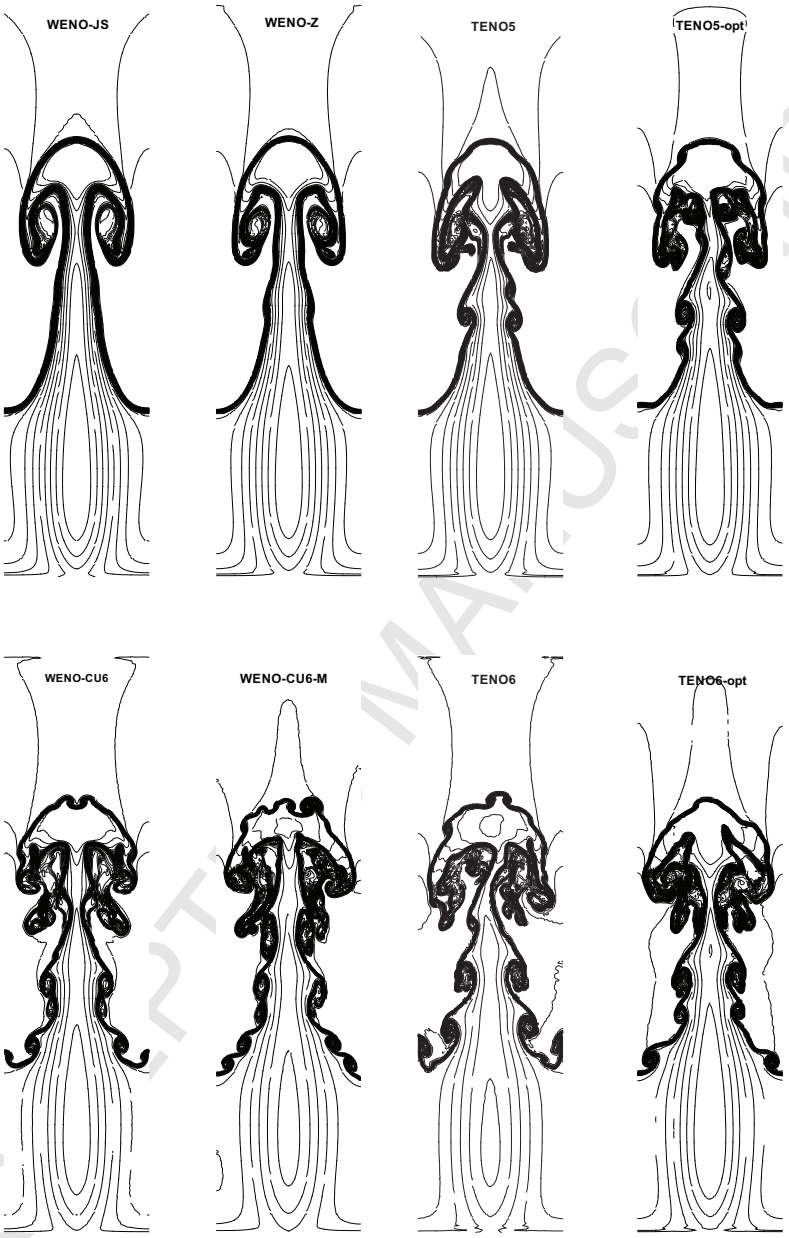


Figure 19: Rayleigh-Taylor instability: density contours. Resolution 128×512.

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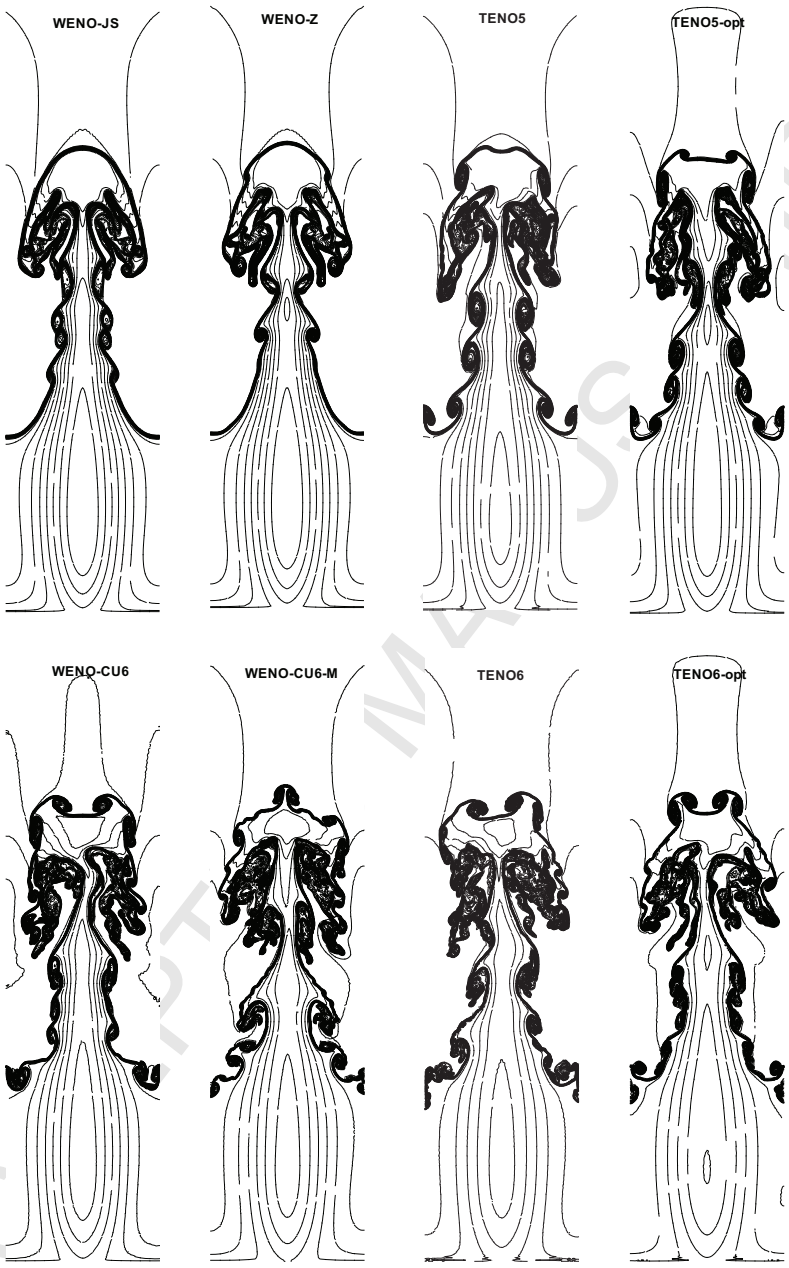


Figure 20: Rayleigh-Taylor instability: density contours. Resolution 256×1024.

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5.7. Two-dimensional viscous shock-tube problem

This case is general for the study of shock/boundary-layer interactions, and has been investigated in [43] [44] with different numerical methods. The initial condition is

$$(\rho, u, v, p) = \begin{cases} (120, 0, 0, 120/\gamma), & \text{if } 0 \leq x < 1/2, \\ (1.2, 0, 0, 1.2/\gamma), & \text{if } 1/2 \leq x \leq 1. \end{cases} \tag{42}$$

The computational domain is $[0, 0.5] \times [0, 1]$. The viscosity is assumed to be constant and the Prandtl number is 0.73. A symmetry condition and an adiabatic viscous wall condition ($\frac{\partial T}{\partial n} = 0$) are imposed for the upper side of the domain and the other boundaries, respectively. The viscous stresses and heat-fluxes are calculated by a generic 4th-order central scheme. Initially, a Mach number 2.37 shock wave, located at x = 1/2, separates two different states of the stationary gas. The incident shock wave and contact discontinuity propagate from left to right, producing a thin boundary layer near the horizontal no-slip wall. After the shock wave reflects at the right wall at time $t = 0.2$, strong shock/boundary-layer interactions develop generating complex flow structures and shock reflections [43]. We simulate this problem at Reynolds number 1000, and the density field at the non-dimensional time $t = 1$ will be discussed. A reference solution is computed utilizing the fifth-order WENO-JS scheme at resolution 5120×2560 as shown in Fig. 21. In order to compare different schemes, we employ a coarse mesh with effective resolution of 1280×640.

As shown in Fig. 22, the results of distinct schemes show remarkable differences, which should be attributed primarily to the different resolution capabilities of the boundary layer and the separation bubbles. Specifically, the

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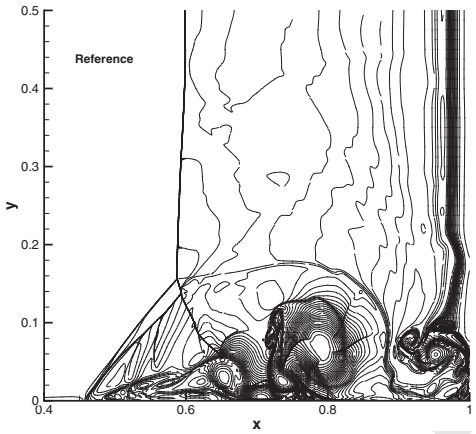


Figure 21: Two-dimensional viscous shock-tube problem: reference density result from WENO-JS scheme. Resolution 5120×2560.

large-vortices shapes and shock-wave patterns produced by WENO-JS and WENO-Z are significantly different from the reference solution and the "converged" result in [44]. TENO5 and TENO5-opt, however, predict such structures quite well. And TENO5 and TENO5-opt have captured the incident shock wave much more sharply than WENO-JS. Furthermore, Fig. 23 reveals that TENO6 and TENO6-opt produce typically much better results than the other schemes. Especially, the dominant vortex structure of TENO6-opt fits the reference quite well whereas that from WENO-CU6-M is slightly distorted due to the lack of dissipation in high wave-number range.

6. Conclusions

In this paper, we have proposed a new concept to construct a family of high-order targeted ENO schemes. Fifth-order and sixth-order TENO schemes are designed and discussed in detail. Extensive one-, two- and three-

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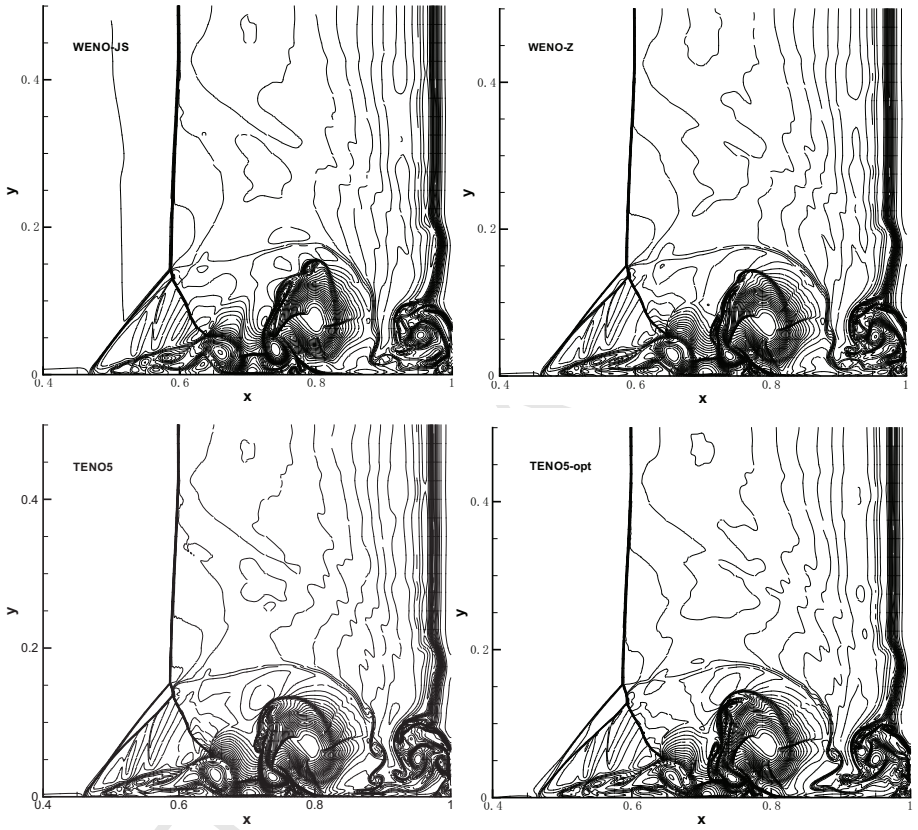


Figure 22: Two-dimensional viscous shock-tube problem: density solutions from five-point stencil schemes. Resolution 1280×640.

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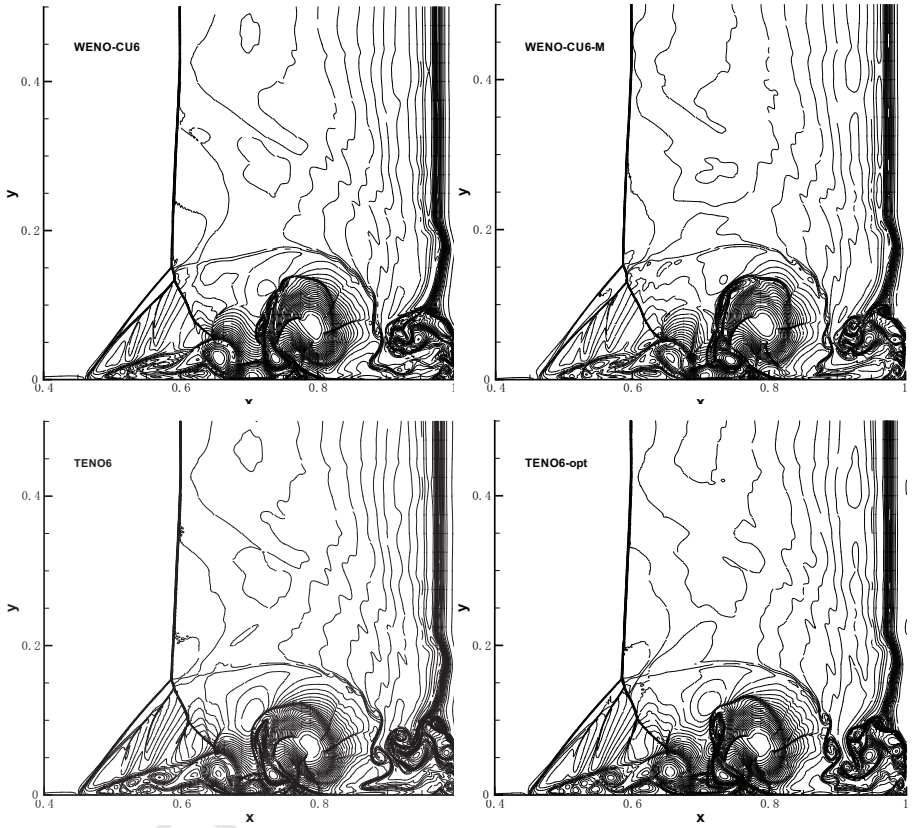


Figure 23: Two-dimensional viscous shock-tube problem: density solutions from six-point stencil schemes. Resolution 1280×640.

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dimensional benchmark tests, involving strong discontinuities and broadband fluctuations, have been performed. Concluding comments about the performances of underlying schemes are as follows:

- Robustness: except for WENO-CU6-M, which fails in the test problem "Interacting blast waves" and "Double Mach reflection of a strong shock", the investigated schemes perform well. In particular, for all cases, the new TENO schemes produce mostly better results than previous solutions with a unique fit of parameters. Note that the positivity-preserving method proposed by Hu and Adams [31] can also be applied to the TENO schemes to further improve robustness.

- Discontinuity capturing: WENO-CU6 and WENO-CU6-M exhibit minor numerical oscillations and overshoots for strong contact discontinuities and shock waves. The new TENO schemes capture such discontinuities slightly sharper than classical WENO-JS in most cases while preserving strict monotonicity.

- Wave-resolution capability: the new TENO schemes clearly resolve intermediate and high wave-number fluctuations better than the other schemes.

- Efficiency: although not specially studied, the arithmetic operations of TENO schemes are similar to that of the classical WENO schemes. The most time-consuming part calculating the smoothness measurements as almost the same.

Classical WENO schemes exhibit robustness problems when there exist multiple shock waves close to each other as all stencils may contain discon-

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tinuities. The TENO schemes address this issue based on a combination of low-order approximation polynomials with incremental stencil width. The combined low-order stencils with narrow width improve the robustness and allow for convenient construction of arbitrarily high-order (both odd- or even-order) schemes.

In order to capture shock waves while resolving broadband fluctuations, strong separation of these entities effectively is necessary. For general WENO schemes, a large value of C improves the performance in smooth regions but degrades the shock capturing capability, thus it is difficult to achieve a sufficient scale separation. With the TENO schemes, this is accomplished by increasing q up to 6 and decreasing C to 1 compared with WENO-CU6-M.

For the TENO schemes, it is fairly simple to control the dissipation in both smooth and non-smooth regions while still remaining sensitive to shock waves by a ENO-like stencil selection. Unlike the classical WENO schemes which favor the selection of the smoother stencils by a convex combination, the stencil is fully suppressed when it is detected as non-smooth according to a threshold. Otherwise it always contributes to the final reconstruction with its standard weight. The threshold is determined by spectral analysis and is problem independent with efficient scale-separation as the priority.

For high-resolution schemes, high-order accuracy is not the most important issue, e.g. high odd-order upwind scheme may still be unnecessarily dissipative. We optimize the TENO schemes by enforcing certain relation between dissipation and dispersion. For order K TENO scheme, this is achieved through adjusting the combination of two order $(K-1)$ component stencils.

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Since the presented TENO5 and TENO6 schemes exhibit good spectral properties and shock-capturing capabilities, it can be expected that eighth- or even higher- order TENO schemes may provide further improved performances.

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AppendixA. Construction of the global reference smoothness indicator τ_K

Besides τ_5 and τ_6 given in the main text, a general τ_K can be constructed on K-point full stencil as

$$\tau_K = \sum_{j=K'}^{K-1} \Delta x^{2j-1} \int_{x_{i-1/2}}^{x_{i+1/2}} \left(\frac{d^j}{dx^j} \hat{f}_K(x) \right)^2 dx, \tag{A.1}$$

where

$$K' = \begin{cases} \frac{K+1}{2}, & \text{if } \mod(K,2) = 1 \\ \\ \frac{K}{2}, & \text{if } \mod(K,2) = 0 \end{cases}. \tag{A.2}$$

This leads to

$$\tau_K = \Delta x^{2K'} (f^{(K')}(x_i))^2 + O(\Delta x^{2K'+2}) \approx O(\Delta x^K). \tag{A.3}$$

It is easy to find that τ_K satisfies the condition Eq. (29) for the formal order of accuracy in smooth regions even near critical points.

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